XX18/4/CHEMI/HP2/ENG/TZ2/XX/M

Markscheme

May 2018

Chemistry

Higher level

Paper 2

22 pages

Question 1

Question		**Answers**	**Notes**	**Total**
1.	**a**	$n(HNO_3) = 0.0500\ dm^3 \times 0.120\ mol\ dm^{-3} = 0.00600$ / 6.00×10^{-3} mol ✓		**1**
1.	**b**	$3HNO_3\ (aq) + Al(OH)_3\ (s) \rightarrow Al(NO_3)_3\ (aq) + 3H_2O\ (l)$ ✓ *Accept ionic equation.*		**1**
1.	**c**	$n(HNO_3) = n(NaOH) = 0.02240\ dm^3 \times 0.1050\ mol\ dm^{-3} = 0.002352$ / 2.352×10^{-3} mol ✓		**1**
1.	**d**	$n(HNO_3)$ reacted = 0.00600 – 0.002352 = 0.003648 / 3.648×10^{-3} mol ✓		**1**
1.	**e**	$n(Al(OH)_3) = (1/3) \times n(HNO_3) = (1/3) \times 0.003648 = 1.216 \times 10^{-3}$ mol ✓ $m(Al(OH)_3) = 1.216 \times 10^{-3}\ mol \times 78.00\ g\ mol^{-1} = 0.0948$ g ✓	*Award [2] for correct final answer.* *Award [1 max] for 0.0948 g.*	**2**
1.	**f**	% $Al(OH)_3$ = 0.0948 g / 1.62 g × 100 = 5.85 % ✓ *Answer must be to 3 significant figures.*		**1**
1.	**g**	to reduce random errors OR to increase precision ✓	*Accept "to check for systematic errors".*	**1**

Question 2

Question		Answers	Notes	Total
2.	**a**	both axes correctly labelled ✓ correct shape of curve starting at origin ✓ $E_{a(catalyst)} < E_{a(without\ catalyst)}$ on x-axis ✓	*M1: Accept "speed" for x-axis label.* *Accept "number of particles" or "frequency" for y-axis.* *M2: Do not accept curve touching x-axis at high energy.* *M3: Ignore shading under curve.*	**3**
2.	**b i**	curve starting from origin with steeper gradient AND reaching same maximum volume ✓		**1**
2.	**b ii**	rate decreases OR slower reaction ✓ propanoic acid partially dissociated/ionized in solution/water OR lower $[H^+]$ ✓	*Accept "weak acid" or "higher pH".*	**2**
2.	**c**	pH converts wide range of $[H^+]$ into simple log scale/numbers OR pH avoids need for exponential notation OR pH converts small numbers into values between 0 and 14 OR pH allows easy comparison of $[H^+]$ values ✓	*Do not accept "pH is between 0 and 14" alone.*	**1**
2.	**d i**	A: CH_3CH_2COOH / propanoic acid **AND** $CH_3CH_2COO^-$ / propanoate ions ✓ B: $CH_3CH_2COO^-$ / propanoate ions ✓	*Penalise "instead of" once.*	**2**
2.	**d ii**	$K_a = 1.34 \times 10^{-5} = [H^+]^2 / 0.10$ OR $[H^+] = 1.158 \times 10^{-3}$ mol dm^{-3} ✓ pH = 2.94 ✓	*Accept [2] for correct answer from wrong Ka.*	**2**
2.	**d iii**	$CH_3CH_2COO^-$ (aq) + H_2O (l) ■ CH_3CH_2COOH (aq) + OH^- (aq) ✓	*Accept equilibrium arrows.*	**1**
2.	**d iv**	less than 7 ✓		**1**
2.	**e i**	SO_2 (g) + H_2O (l) $\rightarrow$ H_2SO_3 (aq) ✓		**1**
2.	**e ii**	H_2SO_3 (aq) + $ZnCO_3$ (s) $\rightarrow$ $ZnSO_3$ (aq) + CO_2 (g) + H_2O (l) ✓		**1**

Question 3

Ques tion		**Answers**	**Notes**	**Tota l**
3.	**a i**	4 levels showing convergence at higher energy ✓		**1**
3.	**a ii**	arrows (pointing down) from n = 3 to n = 2 **AND** n = 4 to n = 2 ✓		**1**
3.	**a iii**	IE = $\Delta E = h\nu = 6.63 \times 10^{-34}$ J s × 3.28×10^{15} s^{-1} = 2.17×10^{-18} J ✓		**1**
3.	**a iv**	$\lambda = c/\nu = 3.00 \times 10^{8}$ m s^{-1} / 3.28×10^{15} s^{-1} = 9.15×10^{-8} m ✓		**1**
3.	**b i**	same number of shells/outer energy level/shielding **AND** nuclear charge/number of protons/Z_{eff} increases causing a stronger pull on the outer electrons ✓		**1**
3.	**b ii**	K^+ 19 protons **AND** Cl^- 17 protons OR K^+ has two more protons ✓ same number of electrons/isoelectronic thus pulled closer together ✓		**2**
3.	**c i**	4s: ↑↓ 3d: ↑↓ ↑↓ ↑↓ ↑↓ ↑ *Note: Cu is [Ar]* $3d^{10}\ 4s^{1}$		**1**
3.	**c ii**	Anode: Cu(s) → Cu^{2+}(aq) + $2e^-$ ✓ Cathode: Cu^{2+}(aq) + $2e^-$ → Cu(s) ✓	*Accept ■ for →.* *Award [1 max] if equations at wrong electrodes.*	**2**
3.	**c iii**	external circuit/wire **AND** from anode to cathode (positive to negative) ✓		**1**
3.	**c iv**	no change in colour ✓	*Do not accept "becomes lighter/darker".*	**1**
3.	**c v**	oxygen / O_2 ✓		**1**
3.	**d**	Transition metals: contain d and s orbitals which are close in energy OR successive ionization energies increase gradually ✓ Alkali metals: second electron removed from much lower energy level OR removal of second electron requires large increase in ionization energy ✓		**2**

Question 4

Question		**Answers**	**Notes**	**Total**
4.	**a**	$ClO_3^-(aq) + 6H^+(aq) + 6Fe^{2+}(aq) \rightarrow Cl^-(aq) + 3H_2O(l) + 6Fe^{3+}(aq)$ ✓	*Accept ■ for →.*	**1**
4.	**b**	n = 6 ✓ $\Delta G^o = -nFE^o$ $E^o = -(-412 \times 10^3) / (6 \times 96\,500) = 0.711$ V ✓	*Award [2] for correct answer.*	**2**
4.	**c**	$E^o(ClO_3^-/Cl^-) = E^o + E^o(Fe^{3+}/Fe^{2+}) = 0.711 + 0.77 = 1.48$ V ✓		**1**

Question 5

Question		Answers	Notes	Total
5.	**a**	bonds broken: 4(C–H) + 2(H–O) / 4(414) + 2(463) / 2582 kJ ✓ bonds made: 3(H–H) + C≡O / 3(436) + 1077 / 2385 kJ ✓ $\Delta H = \Sigma$ BE(broken) – Σ BE(made) = 2582 – 2385 = +197 kJ ✓	*Award [3] for correct final answer.* *Award [2 max] for –197 kJ.*	**3**
5.	**b i**	ΔH^{o}_{f} for any element = 0 by definition OR no energy required to form an element in its stable form from itself ✓		**1**
5.	**b ii**	$\Delta H^{o} = \Sigma\ \Delta H^{o}_{f}$(products) – $\Sigma\ \Delta H^{o}_{f}$(reactants) = –110.5 + 0 – [–74.6 + (–241.8)] = +205.9 kJ ✓		**1**
5.	**b iii**	bond enthalpies are averaged values over similar compounds OR bond enthalpies are not specific to these compounds ✓		**1**
5.	**c**	$\Delta S^{o} = \Sigma S^{o}_{products} - \Sigma S^{o}_{reactants}$ = 198 + 3 × 131 – (186 + 189) = +216 J K^{-1} ✓		**1**
5.	**d**	$\Delta G^{o} = \Delta H^{o} - T\Delta S^{o}$ = 205.9 kJ – 298 K × (216/1000) kJ K^{-1} = +141.5 kJ ✓	*Accept answer consistent with (b)(ii).*	**1**
5.	**e**	$T = \Delta H^{o} / \Delta S^{o}$ = 205 900 J / 216 J K^{-1} T = 953 K ✓	*Do not award if no working shown for T.*	**1**

Question 6

Question		**Answers**	**Notes**	**Total**
6.	**a**	Q: non-equilibrium concentrations **AND** K_c: equilibrium concentrations OR Q: measured at any time **AND** K_c: measured at equilibrium ✓		**1**
6.	**b**	$Q = [SO_3]^2 / [SO_2]^2[O_2]$ $= (2.00)^2 / (2.00)^2(1.00) = 1.00$ ✓ reverse reaction favoured / reaction proceeds to the left **AND** $Q > K_c$ / 1.00 > 0.500 ✓	*Do not award M2 without comparison.*	**2**
6.	**c i**	$[N_2O_2]$ decreases **AND** exothermic thus reverse reaction favoured ✓	*Accept "products decrease" / "equilibrium shifts left".*	**1**
6.	**c ii**	from equilibrium, step 1 $K_c = [N_2O_2] / [NO]^2$ OR $[N_2O_2] = K_c[NO]^2$ ✓ from step 2, rate $= k_2[N_2O_2][O_2]$ rate $= k[NO]^2[O_2]$ ✓	*Award [2] for correct rate expression with working shown.*	**2**
6.	**d**	$T_1 = 293$ K, $T_2 = 303$ K ✓ $\ln(k_2/k_1) = -E_a/R \times (1/T_2 - 1/T_1)$ $\ln(3) = E_a/(8.314) \times (1/293 - 1/303)$ $E_a = 75.8$ kJ mol^{-1} ✓	*Award [2] for correct final answer.*	**2**

Question 7

Question		**Answers**	**Notes**	**Total**
7.	**a i**	polar bonds between H and group 16 element OR difference in electronegativities between H and group 16 element ✓ uneven distribution of charge/electron cloud OR non-linear/bent/V-shaped/angular shape due to lone pairs OR polar bonds/dipoles do not cancel out ✓	*M2: Do not accept "has a dipole moment" without explanation.*	**2**
7.	**a ii**	number of electrons increases ✓ London/dispersion/instantaneous induced dipole-induced dipole forces increase ✓		**2**
7.	**b**	Electron domain geometry: tetrahedral ✓ Molecular geometry: bent/V-shaped/angular ✓ *Note: PH_2^- same geometry as NH_2^-*	*Both marks can be awarded from correct diagrams.* *M1 requires tetrahedral electron domain.*	**2**
7.	**c i**	Structure I: O(1) = 0, O(2) = 0 ✓ Structure II: O(1) = –1, O(2) = +1 ✓	*Award [1] for both correct rows.*	**2**
7.	**c ii**	structure I **AND** no formal charges OR structure I **AND** no charge transfer between atoms ✓		**1**
7.	**d**	O_3 has bond between single and double bond **AND** O_2 has double bond OR O_3 has bond order of 1.5 **AND** O_2 has bond order of 2 OR bond in O_3 is weaker/longer than in O_2 ✓ O_3 requires longer wavelength ✓	*M1: Do not accept "single and double" without context.*	**2**
7.	**e**	CO_2 non-polar weak London/dispersion forces between molecules ✓ SiO_2 network/lattice/3D/giant covalent structure ✓		**2**

Question 8

Question		Answers	Notes	Total
8.	**a**	Physical: equal C–C bond lengths/strengths OR regular hexagon OR all C–C have bond order of 1.5 ✓ Chemical: undergoes substitution reaction more readily than addition OR does not decolourise bromine water OR more stable than expected ✓	*Accept "all bonds intermediate between single and double".*	**2**
8.	**b i**	$3CH_3CH_2CH_2CH_2OH$ (l) + $Cr_2O_7^{2-}$ (aq) + $8H^+$ (aq) → $3CH_3CH_2CH_2CHO$ (aq) + $2Cr^{3+}$ (aq) + $7H_2O$ (l) correct reactants and products ✓ balanced equation ✓		**2**
8.	**b ii**	Aldehyde: by distillation removed from reaction mixture as soon as formed ✓ Carboxylic acid: heat mixture under reflux to achieve complete oxidation to –COOH ✓	*Accept "cleaner" or similar for distillation.*	**2**
8.	**c i**	136 / 2 / 48 / 4 / 16 $C_8H_8O_2$ ✓		**1**
8.	**c ii**	A: C–H in alkanes, alkenes, arenes AND B: C=O in aldehydes, ketones, carboxylic acids and esters ✓		**1**
8.	**c iii**	Any two of: $C_6H_5COOCH_3$ ✓ OR $CH_3COOC_6H_5$ ✓ OR $HCOOCH_2C_6H_5$ ✓	*Do not penalise Kekulé structures for phenyl group.* *Award [1 max] for two correct aliphatic esters.*	**2**
8.	**c iv**	$C_6H_5COOCH_3$ signal at ~4 ppm due to alkyl group on ester ✓		**1**

Question 9

Question		Answers	Notes	Total
9.	**a i**	$CH_3CH_2COCH_2CH_2CH_3$ ✓ AND $CH_3CH_2CH_2COCH_2CH_3$ ✓ *(hexan-3-one and hexan-2-one... accept any two correct chain ketones)*	*Accept condensed formulas.*	**2**
9.	**a ii**	A: $CH_3CH_2COCH_2CH_2CH_3$ peak at 43 or 71 due to respective fragments ✓ B: $CH_3CH_2CH_2COCH_2CH_3$ peak at 29 or 57 due to respective fragments ✓	*Penalise missing "+" sign once only.*	**2**
9.	**b i**	heterolytic / heterolysis ✓		**1**
9.	**b ii**	polar protic ✓		**1**
9.	**b iii**	carbocation / $R_1R_2R_3C^+$ ✓ Shape: triangular / trigonal planar ✓		**2**
9.	**b iv**	around 50 % each OR similar/equal percentages ✓ nucleophile can attack from either side of the planar carbocation ✓	*Accept "racemic mixture/racemate".*	**2**
9.	**c**	Stage one: $C_6H_5NO_2$ (l) + 3Sn (s) + $7H^+$ (aq) → $C_6H_5NH_3^+$ (aq) + $3Sn^{2+}$ (aq) + $2H_2O$ (l) ✓ Stage two: $C_6H_5NH_3^+$ (aq) + OH^- (aq) → $C_6H_5NH_2$ (l) + H_2O (l) ✓		**2**